



Instructions:

1. Complete Case 1 & 2 according to the guidelines and requirements provided.
2. Submit your solutions within **four (4) calendar days** of receiving this request.
3. This assessment is strictly confidential and must not be shared or discussed publicly or privately with any unauthorized individuals.

Name: Samuel Wong

Email: Samuelw7557@gmail.com

School: Singapore Institute of Technology

Course: BEng (Hons) in Mechatronics Systems

GitHub Repository: https://github.com/Samuelsw1/Rapsodo_QA_Assignment

Submission Date: 14/4/2025

Case-1 Registration Page

The Registration Page enables users and organizations to register and gain access to the system. Write your test cases based on the register page and the related requirements below: PS. You can use any test case/document format.

The image displays the Rapsodo registration page and a detailed breakdown of its form fields. The main page on the left features the Rapsodo logo, a user icon, and input fields for First Name, Last Name, Email, Password, Password Confirmation, Country, State (Choose State), Organization Type (Choose Organization Type), Organization Name, and a date field (mm/dd/yyyy). A red 'CONTINUE' button and a 'Terms & Conditions' checkbox are at the bottom. The right side shows a detailed view of the 'Country', 'Date of Birth', 'State', and 'Organization Type' fields. The 'Country' field lists countries like Afghanistan, Albania, Algeria, American Samoa, Andorra, and Angola. The 'Date of Birth' field shows a calendar for July 2002, with the 28th selected. The 'State' field lists states like Alabama, Alaska, American Samoa, Arizona, Arkansas, California, Colorado, and Connecticut. The 'Organization Type' field lists options like College, Academy, Individual, and Other.

Rapsodo
measure to master

Country:

Country

Country

Afghanistan

Albania

Algeria

American Samoa

Andorra

Angola

Date of Birth:

Pass

Country

Choose State

Choose Organization Type

Organization Name

mm/dd/yyyy

CONTINUE

☐ Terms & Conditions

State:

United States

Choose State

Choose State

Alabama

Alaska

American Samoa

Arizona

Arkansas

California

Colorado

Connecticut

Organization Type:

Choose Organization Type

Choose Organization Type

College

Academy

Individual

Other

Components

- “Profile Photo” field’s supported files are only jpeg, jpg and png. And maximum size should be 10mb. (Unexpected file format. Supported file type jpeg, jpg and png.) (File size too large. Files cannot be larger than 10MB.)

Test Case Field	Details
Test Case ID	TC_UI_1
Test Priority	High
Test Designed by	Samuel Wong
Date of Test Designed	12-Apr-2025
Test Executed by	To be determined
Date of the Test Execution	To be determined
Name or Test Title	Validate Profile Photo Upload Format and Size
Description/Summary of Test	Ensure only JPEG, JPG, PNG files up to 10MB are accepted in the Profile Photo upload field.
Pre-condition	User is on the Registration Page.
Dependencies	File system access, file picker UI component.
Test Steps	1. Open registration form. 2. Click the "Profile Photo" field. 3. Upload an invalid format (e.g., PDF). 4. Upload a valid PNG over 10MB. 5. Upload a valid JPEG under 10MB.
Test Data	1. invalid_file.pdf (5MB) 2. large_image.png (12MB) 3. profile_photo.jpg (2MB)
Expected Results	- Step 3: Error message "Unexpected file format. Supported file types: jpeg, jpg, png." - Step 4: Error message "File size too large. Files cannot be larger than 10MB." - Step 5: File is accepted and preview appears.
Post-Condition	Valid photo is uploaded and visible in preview (if accepted).
Actual Result	<i>(Leave blank until executed)</i>
Status (Fail/Pass)	<i>(Mark after execution)</i>

- “First Name” and “Last Name” fields accept up to 50 characters. Special characters and numbers are not allowed. “First Name” and “Last Name” fields are required.

Test Case Field	Details
Test Case ID	TC_UI_2
Test Priority	High
Test Designed by	Samuel Wong
Date of Test Designed	12-Apr-2025
Test Executed by	To be determined
Date of the Test Execution	To be determined
Name or Test Title	Validate First Name and Last Name Field
Description/Summary of Test	Ensure name fields reject numbers, special characters, and inputs longer than 50 characters.
Pre-condition	User is on the Registration Page.
Dependencies	Registration form, client-side input validation scripts.
Test Steps	1. Leave both fields empty and attempt to continue. 2. Enter names with numbers (e.g., "Sam123"). 3. Enter names with special characters (e.g., "J@ne"). 4. Enter names over 50 characters. 5. Enter valid names (e.g., "Samuel", "Tan").
Test Data	- Empty fields - Sam123, J@ne - A * 51 (51 characters) - Samue1, Tan
Expected Results	- Step 1: Error "First Name is required." & "Last Name is required." - Step 2–3: Error "Special characters and numbers are not allowed." - Step 4: Error "Maximum 50 characters allowed." - Step 5: Fields accept input with no error.
Post-Condition	Valid name entries retained in the form.
Actual Result	<i>(To be filled after execution)</i>
Status (Fail/Pass)	<i>(Mark after execution)</i>

- “Email” field input must be a valid e-mail address. “Email” field is required.

Test Case Field	Details
Test Case ID	TC_UI_3
Test Priority	High
Test Designed by	Samuel Wong
Date of Test Designed	12-Apr-2025
Test Executed by	To be determined
Date of the Test Execution	To be determined
Name or Test Title	Validate Email Field
Description/Summary of Test	Verify that the email field accepts only valid email formats and that it is a required field.
Pre-condition	User is on the Registration Page.
Dependencies	Form validation logic (regex-based), server-side validation rules.
Test Steps	1. Leave the field empty and attempt to continue. 2. Enter an invalid format: user . com 3. Enter an invalid format: @gmail . com 4. Enter an invalid format: user@ . com 5. Enter a valid email: samuel@example . com
Test Data	- (blank) - user . com - @gmail . com - user@ . com - samuel@example . com
Expected Results	- Step 1: Error “Email is required.” - Steps 2–4: Error “Invalid email format.” - Step 5: Input accepted with no error.
Post-Condition	Valid email is retained in the field.
Actual Result	<i>(To be filled after execution)</i>
Status (Fail/Pass)	<i>(Mark after execution)</i>

- “Password” must have at least 8 characters and includes at least one uppercase letter, one lowercase letter and one number. “Password” and “Password Confirmation” fields are required.

Test Case Field	Details
Test Case ID	TC_UI_4
Test Priority	High
Test Designed by	Samuel Wong
Date of Test Designed	12-Apr-2025
Test Executed by	To be determined
Date of the Test Execution	To be determined
Name or Test Title	Validate Password Field
Description/Summary of Test	Ensure the password field enforces the minimum complexity: 8+ characters, 1 uppercase, 1 lowercase, 1 digit.
Pre-condition	User is on the Registration Page
Dependencies	Password validation logic (client-side and/or server-side)
Test Steps	1. Leave field empty 2. Enter "test" 3. Enter "password" 4. Enter "Password" 5. Enter "Password1"
Test Data	- (blank) - test - password - Password - Password1
Expected Results	- Step 1: "Password is required" - Step 2-4: "Password does not meet complexity" - Step 5: Password accepted
Post-Condition	Password is accepted only if it passes all validation checks
Actual Result	(To be filled after execution)
Status (Fail/Pass)	(Mark after execution)

Test Case Field	Details
Test Case ID	TC_UI_5
Test Priority	High
Test Designed by	Samuel Wong
Date of Test Designed	12-Apr-2025
Test Executed by	To be determined
Date of the Test Execution	To be determined
Name or Test Title	Validate Password Confirmation Match
Description/Summary of Test	Ensure that the "Password Confirmation" field matches the "Password" field exactly.
Pre-condition	User has entered a valid password
Dependencies	Password confirmation field logic
Test Steps	1. Enter valid password: Password1 2. Leave confirmation empty 3. Enter mismatched password: Password2 4. Enter matching password: Password1
Test Data	Password = Password1 Confirm Password = "", Password2, Password1
Expected Results	- Step 2: Error – "Password confirmation is required" Step 3: Error – "Passwords do not match" Step 4: Success
Post-Condition	Proceed only when both password and confirmation match
Actual Result	(To be filled after execution)
Status (Fail/Pass)	(Mark after execution)

- The “Country” field is a combobox in which countries are listed. Only if “United States” is selected from the country combobox, the “State” combobox becomes enabled. Otherwise “State” field is disabled and not to be selectable. “Country selection is mandatory, state is not.

Test Case Field	Details
Test Case ID	TC_UI_6
Test Priority	Medium
Test Designed by	Samuel Wong
Date of Test Designed	12-Apr-2025
Test Executed by	To be determined
Date of the Test Execution	To be determined
Name or Test Title	Validate Country and State ComboBox Logic
Description/Summary of Test	Ensure that the “State” combo box is only enabled when “United States” is selected in the Country field.
Pre-condition	Page is loaded with all dropdowns visible
Dependencies	Dynamic UI rendering of state field based on country selection
Test Steps	1. Open the Country dropdown 2. Select “Singapore” → Verify “State” is disabled 3. Select “United States” → Verify “State” is enabled
Test Data	Country options: “Singapore”, “United States”
Expected Results	- Step 2: State field remains disabled - Step 3: State field becomes enabled
Post-Condition	UI reflects expected interactivity logic
Actual Result	<i>(To be filled after execution)</i>
Status (Fail/Pass)	<i>(Mark after execution)</i>

- There are 4 types of organization; “College”, “Academy”, “Individual” and “Other”. Only 1 organization can be selected from the “Organization Type” field. Organization type selection is mandatory.

Test Case Field	Details
Test Case ID	TC_UI_7
Test Priority	High
Test Designed by	Samuel Wong
Date of Test Designed	12-Apr-2025
Test Executed by	To be determined
Date of the Test Execution	To be determined
Name or Test Title	Validate Mandatory and Single Selection for “Organization Type”
Description/Summary of Test	Ensure the “Organization Type” field requires a selection and only allows one selection at a time.
Pre-condition	Registration form is open and all organization types are visible
Dependencies	“Organization Type” combobox must be functioning
Test Steps	1. Try submitting the form without selecting any organization type 2. Select “College”, then try to select “Academy” simultaneously
Test Data	Org types: College, Academy, Individual, Other
Expected Results	- Step 1: System should display an error that organization type is required - Step 2: Only one organization type can remain selected
Post-Condition	Form does not submit unless one organization type is selected
Actual Result	<i>(To be filled after execution)</i>
Status (Fail/Pass)	<i>(Mark after execution)</i>

- Only if “Other” is selected from the organization type combobox, the “Organisation Name” text box becomes enabled. Otherwise “Organization Name” field is disabled and not to be editable. “Organization Name” fields accept up to 100 characters. Special characters and numbers are allowed. Organization name field is not required.

Test Case Field	Details
Test Case ID	TC_UI_8
Test Priority	Medium
Test Designed by	Samuel Wong
Date of Test Designed	12-Apr-2025
Test Executed by	To be determined
Date of the Test Execution	To be determined
Name or Test Title	Validate Conditional Activation of “Organization Name” Field
Description/Summary of Test	Ensure that the “Organization Name” text box only becomes editable when “Other” is selected.
Pre-condition	Registration form is loaded with all organization types visible
Dependencies	UI logic tied to organization type selection
Test Steps	1. Open Organization Type dropdown 2. Select “College”, “Academy”, or “Individual” → Check field 3. Select “Other” → Check field
Test Data	Org types: College, Academy, Individual, Other
Expected Results	- Step 2: “Organization Name” is disabled and non-editable - Step 3: “Organization Name” is enabled and editable
Post-Condition	Organization Name field reflects correct enabled/disabled state
Actual Result	<i>(To be filled after execution)</i>
Status (Fail/Pass)	<i>(Mark after execution)</i>
Notes	Field accepts up to 100 characters including special characters and numbers when enabled

- Birthdate; the calendar shows 18 years earlier than the current day by default. Under the age of 18 cannot be a member of the system. Date selection is mandatory.

Test Case Field	Details
Test Case ID	TC_UI_9
Test Priority	High
Test Designed by	Samuel Wong
Date of Test Designed	12-Apr-2025
Test Executed by	To be determined
Date of the Test Execution	To be determined
Name or Test Title	Validate Birthdate Age Restriction and Mandatory Selection
Description/Summary of Test	Verify the birthdate calendar shows default date 18 years prior, restricts underage users, and is required.
Pre-condition	Registration form is open
Dependencies	Datepicker UI must be functional
Test Steps	1. Open the date picker 2. Check the default date 3. Select a date that makes user under 18 4. Leave the field empty and submit
Test Data	Dates like 2010-04-12 (underage), blank field
Expected Results	- Step 2: Default date should be today's date minus 18 years - Step 3: Error shown: "You must be 18 or older" - Step 4: Error shown: "Birthdate is required"
Post-Condition	Form does not submit unless valid birthdate is selected
Actual Result	<i>(To be filled after execution)</i>
Status (Fail/Pass)	<i>(Mark after execution)</i>
Notes	Needs proper validation at both frontend (UI) and backend

- Terms & Conditions is a link that opens the “terms of use” URL. Checkbox is required.

Test Case Field	Details
Test Case ID	TC_UI_10
Test Priority	High
Test Designed by	Samuel Wong
Date of Test Designed	12-Apr-2025
Test Executed by	To be determined
Date of the Test Execution	To be determined
Name or Test Title	Validate Terms & Conditions Link and Required Checkbox
Description/Summary of Test	Ensure the Terms & Conditions link opens correctly and the checkbox is required before submission.
Pre-condition	Registration form is open
Dependencies	Link must be functional; checkbox validation implemented
Test Steps	1. Click on the Terms & Conditions link 2. Leave the checkbox unchecked and click “Continue” 3. Tick the checkbox and submit again
Test Data	N/A
Expected Results	- Step 1: A new tab/window opens with the correct Terms of Use URL - Step 2: Error shown: “You must agree to the terms and conditions” - Step 3: Form proceeds successfully (if other validations pass)
Post-Condition	User must accept T&Cs to register
Actual Result	<i>(To be filled after execution)</i>
Status (Fail/Pass)	<i>(Mark after execution)</i>
Notes	Test both hyperlink behavior and checkbox validation

- After clicking the “CONTINUE” button, the user will be redirected to the success page and a verification link will be sent to the entered email account.



Test Case Field	Details
Test Case ID	TC_UI_11
Test Priority	High
Test Designed by	Samuel Wong
Date of Test Designed	12-Apr-2025
Test Executed by	To be determined
Date of the Test Execution	To be determined
Name or Test Title	Redirect and Email Verification After Clicking Continue
Description/Summary of Test	Verify users are redirected to the success page and an email verification link is sent upon valid form submission.
Pre-condition	All required form fields must be valid and filled in
Dependencies	Working email backend, success page must exist
Test Steps	1. Fill all required fields with valid data 2. Click “Continue” button
Test Data	Valid user inputs across all fields including a valid email address
Expected Results	- Step 1: Form submission completes - Step 2: User is redirected to success/confirmation page - Step 3: A verification link is sent to the provided email
Post-Condition	User must check email to verify and activate account
Actual Result	(To be filled after execution)
Status (Fail/Pass)	(Mark after execution)

Case-2 Rapsodo.com - Adding Product To The Cart

With using any test automation framework please do the following;

1- The user goes to the "https://rapsodo.com" address and verifies that they went to this address. 2- The user clicks on the "Cart" icon at the top right area.

The user verifies that the shopping cart is empty at that moment.

3.- The user clicks on the "Golf" button and chooses the "Mobile Launch Monitor (MLM)" via "Products" button.

Press "Shop MLM" button. The user asserts / verifies that the page's title is " **Mobile Launch Monitor (MLM)**".

Select "**MOBILE LAUNCH MONITOR**" from the right side of the web site.

4- The user clicks on the "ADD TO CART" button

Verifies that the user redirected to the cart screen.

Verify the price of the **Mobile Launch Monitor (MLM)** is the same with **Mobile Launch Monitor (MLM)** screen

5- The user increases quantity (quantity = 2) of product on the "cart".

The user verifies that there are two items in their "My Cart" part and that it costs \$xx.xxx.00.

Please share the URL of your *private* GitHub repository by replying to the same email in which you received this technical assignment.

```

PS C:\Rapsodo\chrome-win64> python Case_2.py

DevTools listening on ws://127.0.0.1:60631/devtools/browser/a0f7ab3b-7c32-4a6f-8a1f-79b3b9042f45
🚀 Starting Case 2 test...
🌐 Opened https://rapsodo.com
🔍 Looking for cart icon...
✅ Clicked cart icon and navigated to cart.
✅ Verified cart is empty
⚠️ Clicking 'Golf' link...
✅ Navigated to Golf page
[24088:22380:0413/204815.739:ERROR:device_event_log_impl.cc(202)] [20:48:15.739] USB: usb_service_win.cc:105 SetupDiGetDeviceProperty({{A45C254E-DF1C-4EFD-8
020-67D146A858E0}, 6}) failed: Element not found. (0x490)
🔍 Looking for 'Products' button under Golf...
✅ Clicked 'Products'
🔍 Looking for 'Shop MLM' button...
✅ Clicked 'Shop MLM' button
📄 Verified page title contains 'Mobile Launch Monitor'
Created TensorFlow Lite XNNPACK delegate for CPU.
Attempting to use a delegate that only supports static-sized tensors with a graph that has dynamic-sized tensors (tensor#-1 is a dynamic-sized tensor).
✅ Selected correct 'Mobile Launch Monitor (MLM)' variant at $299.99
❌ Corrected hidden input to ensure $299.99 variant is added.
📄 Submitting correct variant to cart
✅ Submitted correct variant directly to cart using fetch()
📄 Returned to cart page
📄 Cart price: $249.99
📄 Quantity updated to 2
✅ Verified cart shows quantity 2
✅ Total price for 2 items: $499.98
📄 Screenshot saved to Desktop
✅ Test finished.

```

Requirements

- **Python 3.10+**
- **Selenium** installed (pip install selenium)
- **Google Chrome** (latest version)
- **ChromeDriver** matching Chrome version
(Download: chromedriver.chromium.org)

https://github.com/Samuelsw1/Rapsodo_QA_Assignment